MARK SCHEME · TEACHER COPY

12BR · Topic 16: System Software

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Total marks: 72

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Question 1 · Oct/Nov 2025 Paper 31, original Q6 [4 marks]

Question	Answer	Marks
1	**One** mark for each mark point (**Max 4**) MP1 The interpreter translates the source code one line at a time MP2 If the line is syntax error free it is executed MP3 It is not stored in executable format MP4 If an error is found, the program halts with an error message MP5 Each line must be translated every time it is run, including lines running multiple times for example in loops	4

Question 2 · Oct/Nov 2025 Paper 31, original Q7 (a),(b) [6 marks]

Question	Answer	Marks
2(a)	**One** mark for each correct answer **#Jd7** – must begin with a member of the group uppercase // cannot begin with a symbol **C%6A** – the fourth character cannot be a member of the group uppercase // the fourth character must be either a symbol, digit or lowercase	2
2(b)	**One** mark **per** mark point <uppercase> ::= A | C | E | G | J <passcode> ::= <uppercase><code> <code> ::= <lowercase>|<symbol>|<digit> |<lowercase><code>|<symbol><code>|<digit><code>	4

Question 3 · Oct/Nov 2025 Paper 31, original Q8 (a),(b) [6 marks]

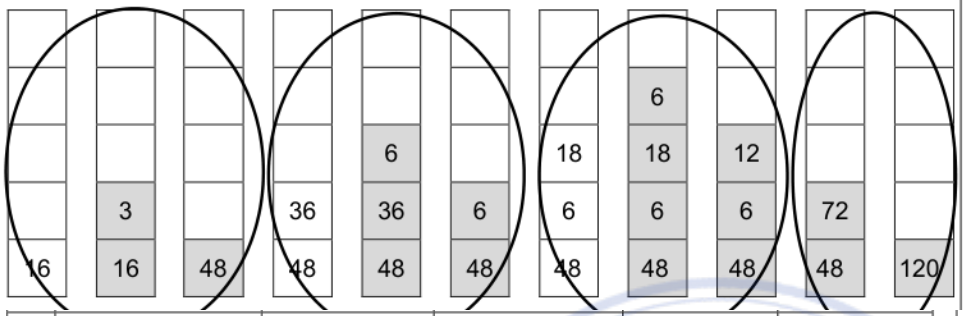
Question	Answer	Marks
3(a)	**One** mark for each mark point MP1 Running multiple processes concurrently MP2 … which benefits process management by allowing more tasks to complete than would be the case if they had to run one task after another.	2

3(b)	**One** mark for each mark point (**Max 3**) MP1 The processes are queued as they arrive MP2 Processes with the shortest burst time are executed first MP3 It is a pre-emptive scheduling function // When a process with a shorter burst time arrives the existing process is replaced by the shorter process. MP4 The scheduler will continue to choose shorter processes over longer processes if they continue to be added to the queue can cause starvation for longer jobs **One** mark for benefit (**Max 1**) e.g. MP5 Processes with a short burst time are processed very quickly MP6 Waiting time is minimised	**4**

Question 4 · May/June 2026 Paper 33, original Q7 (a),(b),(c) [9 marks]

Question	Answer	Marks	Guidance
4(a)	**One** mark per mark point (**Max 3**) MP1 The RPN expression is read from left to right MP2 If the next element is a number it is pushed onto the stack MP3 If the next element is an operator, the last two elements are popped from the stack MP4 …the operator is applied to them and the result is pushed onto the stack MP5 The process repeats until the whole expression has been evaluated	**3**	
4(b)	**One** mark per mark point (**Max 2**) MP1 `18 15 - 3 +` MP2 `5 * 12 /` Final correct expression `18 15 - 3 + 5 * 12 /`	**2**	

Total: 72 marks

4(c)	**One** mark per ring (**Max 4**)  	**4**	Values in shaded areas are necessary to award the marks in the rings.

4	W⇨N2⇨N5	N6	4+7+4=15	3	18

Question 5 · May/June 2026 Paper 32, original Q6 (a),(b),(c) [9 marks]

Question	Answer	Marks	Guidance
5	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.		

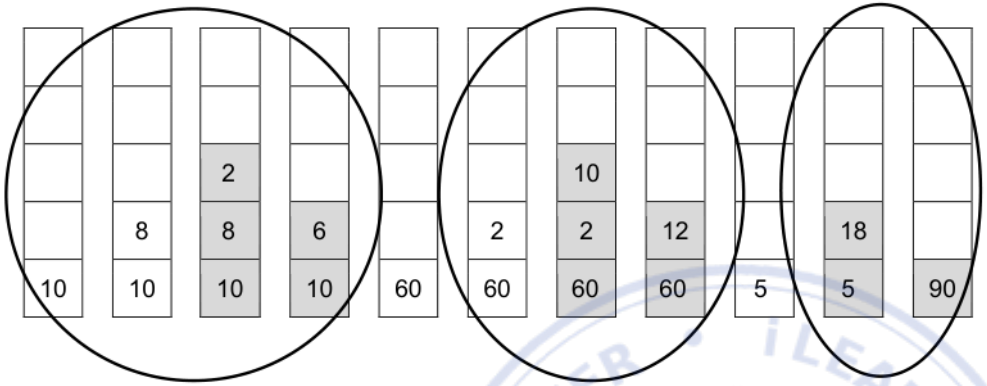
Annotation	Meaning
✓	Correct
✗	Incorrect
λ	To indicate where a key word/phrase/code is missing.
~	Not relevant or used to separate parts of an answer.
~~	Indicates a part of the answer that is incorrect.
Highlighter	To draw attention to a particular aspect or to indicate where parts of an answer have been combined.
TV	Too vague.
REP	Repetition

NE	No examples or not enough.

Annotation	**Meaning**
BOD	Benefit of Doubt.
NAQ	Not Answered Question.
SEEN	Indicates that work or a page has been seen including blank answer spaces and blank pages.
FT	Follow through.
I	Ignore

5(a)	**One** mark per mark point (**Max 2**) MP1 `16 9 7 -` MP2 `18 6 + / +` Final correct expression `16 9 7 - 18 6 + / +`	**2**	

5(b)	**One** mark per mark point (**Max 3**)
	MP1 `(6 + 15) // 6 + 15`
	MP2 `- (20 - 3)`
	MP3 `/ (27 * 9)` **and** the pair of brackets around the whole numerator
	
	Final correct expression
	`((6 + 15) - (20 - 3)) / (27 * 9) //`
	`(6 + 15 - (20 - 3)) / (27 * 9)`
		3	

5(c)	**One** mark per ring (**Max 3**) **One** mark for the interim totals (60) **and** (5).  	**4**	Do not allow operators in stacks Do not allow letters in stacks Values in shaded areas are necessary to award the marks in the rings.

Question 6 · Oct/Nov 2025 Paper 32, original Q5 (a),(b) [7 marks]

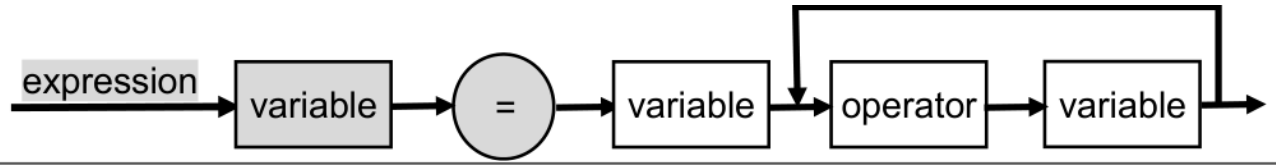
Question	Answer	Marks
6(a)	**One** mark per point • Running state • Ready state • Blocked state	**3**

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6(b)	**One** mark per mark point (**Max 3**) MP1 The processes are queued as they arrive MP2 The process with the shortest time to complete/burst time is selected first and executed MP3 The process will continue until complete or put in a waiting state once execution has begun // it is non-pre-emptive MP4 The scheduler will continue to choose shorter processes over longer processes if they continue to be added to the queue can cause starvation for longer jobs **One** mark for a benefit (**Max 1**) MP5 Shorter jobs don't have to wait for longer jobs to complete before processing // Significantly reduces the average overall waiting time for processes // Ensures starvation doesn't occur for process with shorter burst times MP6 Higher throughput of processes	**4**

Question 7 · Oct/Nov 2025 Paper 32, original Q9 (a),(b),(c),(d) [6 marks]

Question	Answer	Marks
7(a)	Variables must begin with a letter / not a digit	**1**
7(b)	`<operator> ::= + | - | * | / | ^`	**1**
7(c)	**One** mark per mark point MP1 Two variable boxes added to diagram MP2 One operator box added to diagram and all boxes in correct order MP3 Connections, arrows and return loop correctly added and no additional boxes or connections	**3**

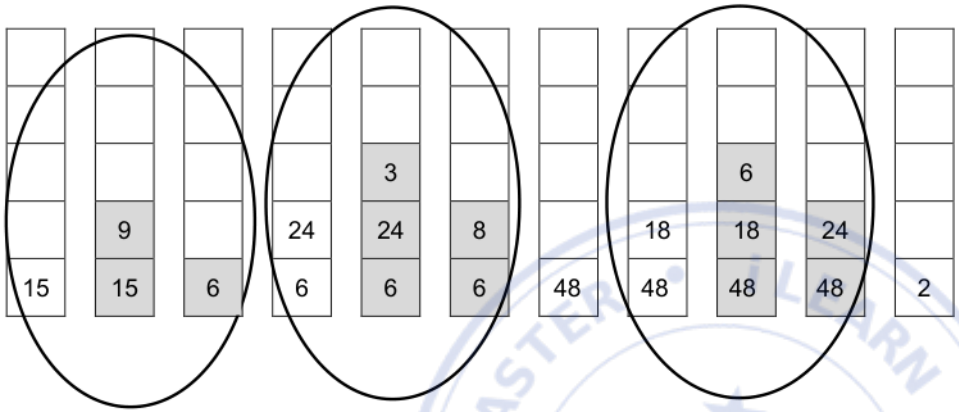
		
7(d)	Answer must begin with a valid letter. It can then be followed by any number of valid digits and/or letters, as long as it is at least four characters in length. Example answer AC768	**1**

Question 8 · May/June 2026 Paper 31, original Q7 (a),(b),(c) [9 marks]

Question	Answer	Marks	Guidance
8(a)	**One** mark per mark point (**Max 2**) MP1 `21 15 3 -` MP2 `8 + * 7 /` Final correct expression `21 15 3 - 8 + * 7 /`	**2**	
8(b)	**One** mark per mark point (**Max 3**) MP1 `(c + d)` MP2 `(a - b) * (a - c)` MP3 `/ (` MP2 `)` Final correct expression `(c + d) / ((a - b) * (a - c))`	**3**	

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Total: 72 marks

8(c)	**One** mark per ring (**Max 3**) **One** mark for the interim total (48) **and** the final total (2).  	**4**	Do not allow operators in stacks Values in shaded areas are necessary to award the marks in the rings.

Question 9 · Oct/Nov 2025 Paper 33, original Q6 (a),(b) [6 marks]

Question	Answer	Marks
9(a)	**One** mark for each mark point (**Max 2**) • Process scheduling is required to ensure that all processes are executed in a timely manner • … and enables multitasking/multiprogramming/multiprocessing • … to minimise CPU idle time • … to ensure that no process is starved of resources. // … to ensure fair access to resources. • … ensures jobs/processes are completed in order of priority.	**2**

Total: 72 marks

9(b)	**One** mark for each mark point (**Max 3**) MP1 Processes are queued as they arrive. MP2 It is a pre-emptive scheduling routine. MP3 A fixed time quantum is given to each process. // Each process has an equal time slice. MP4 When a time slice ends, the status of the process is saved/queued so it can continue from where it left off in its next time slice. MP5 and the next process is executed for its time slice; its previous state is reinstated/restored, if applicable. MP6 If a process completes within its time slice, the next process is executed for its time slice. **One** mark for benefit (**Max 1**) MP7 Reduces average response time by limiting each process to a fixed amount of time. MP8 No issue with starvation of resources.	**4**

9							19				

Question 10 · Oct/Nov 2025 Paper 33, original Q8 [4 marks]

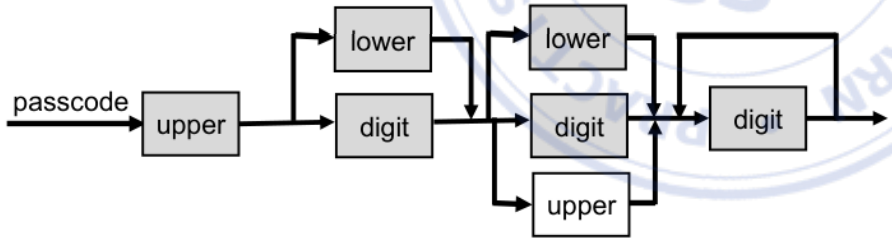
Question	Answer	Marks
10	**One** mark per mark point (**Max 4**) MP1 It is the second stage of compilation // It's the compilation stage after lexical analysis. MP2 It takes input from the lexical analyser in the form of token streams. MP3 The source code is analysed / parsed against the rules of the language to detect any errors in the code. MP4 The output from this phase is a parse tree. MP5 Syntax errors are reported.	**4**

10								0			

Question 11 · Oct/Nov 2025 Paper 33, original Q9 (a),(b),(c) [6 marks]

Question	Answer	Marks
11(a)	The second character must come from either lower or digit. It can't be upper.	**1**

11(b)	**Max 3**
	
	One mark
	MP1 `<upper> ::= J | K | L | V | X | Z`
	
	Either: fully written out answer
	MP2 Any two correct options for `<passcode>`
	MP3 Remaining two options correct for `<passcode>`
	
	Example answer
	`<passcode> ::=`
	`<upper><lower><lower><digit>|`
	`<upper><lower><digit><digit>|`
	`<upper><digit><digit><digit>|`
	`<upper><digit><lower><digit>`
	
	Or: answer with interim expression
	MP4 `<passcode> ::= <upper><middle><middle><digit>`
	MP5 `<middle> ::= <lower>|<digit>`
		3

11(c)	**One** mark per mark point MP1 Box for upper in correct place with correct connections MP2 Correct repetition arrow for final digit 	2
11		

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